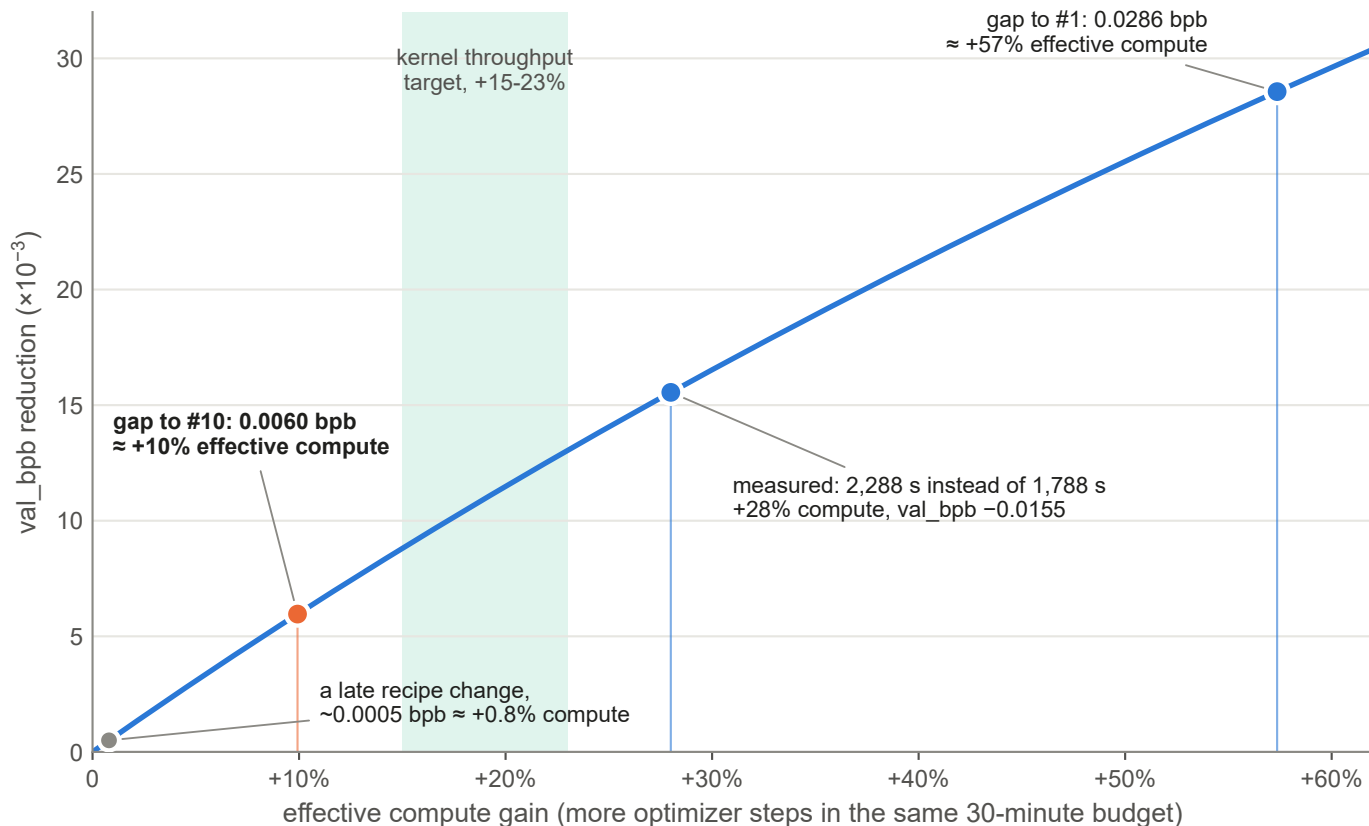


What extra compute is worth

Near our operating point, val_bpb falls with the log of effective compute: $\Delta\text{bpb} \approx 0.063 \cdot \ln(\text{compute ratio})$.



The curve is fitted on one +28% run (24 Sep). Its slope near zero is about 0.00063 bpb per 1%; the per-step rate measured directly on two chips is about 0.00057 bpb per 1% more steps. Gaps are from our best official score, 0.9614 (K82s4), to the final leaderboard snapshot, by rank only (docs/GAP-ANALYSIS.md).